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.....//.....//.....

(Q1)

- Complexity

  Enumeration → all possible combinations for all the variables are calculated to see wanted result. It's a brute force algorithm.

  Variable Elimination → eliminates unnecessary variable on each step. with

  → Variable Elimination is always faster than Enumeration

- Reuse of Computation

  Enumeration → Never holds a intermediate factor, so similar combinations calculated over and over

  Variable Elimination → Creates a intermediate factor after every elimination step. Those factor are reusable for future calculations.

  → Variable Eliminations creates reusable factors.

- Scaleability

  Enumeration → Because of the exponential complexity, it scale bad.

  Variable Elimination → Takes advantages of network connection, so only calculate relevant content. That makes VE good scale

  → Variable Elimination scales much better than Enumeration

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Q2)

$P(A=1) = 0.3$
$P(A=0) = 0.2$

- $P(B=1 \mid A=1) = 0.8 = \dfrac{P(B=1 \cap A=1)}{P(A=1)}$

$$0.8 \times 0.3 = P(B=1 \cap A=1) = 0.24$$

- $P(B=1 \mid A=0) = 0.1 = \dfrac{P(B=1 \cap A=0)}{P(A=0)}$

$$0.1 \times 0.7 = P(B=1 \cap A=0) = 0.07$$

$$\rightarrow P(B=1 \cap A=0) + P(B=1 \cap A=1) = P(B=1)$$
$$0.7 + 0.24 = 0.31$$

$$\text{?} \quad P(A=1 \mid B=1) = \frac{P(A=1 \cap B=1)}{P(B=1)} = \frac{0.24}{0.31}$$

$$= 0.77$$

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Q3

$A \rightarrow B \rightarrow C \quad ; \quad C = 1$

$\lambda_C(C=1) = 1$
$\lambda_C(C=0) = 0$
} prior(A)

$$\lambda_B(b) = \sum_c P(c|b)\,\lambda_C(c) = P(C=1 \mid B=b)$$

$$\lambda_A(a) = \sum_b (b|a)\,\lambda_B(a)$$

- Because the evidence is $C=1$ message direction is below to up

$C \rightarrow B \rightarrow A$ (λ messages)

C starts message (evidence node)

B updated with λ_C (B updated first)

C updated with λ_B

- In polytree structure each piece of evidence reaches a node in a single path. So each λ and π messages are independent and complete. Multiplying incoming messages gives the posterior